

西方经济学

Part 2 Classical Theory

THE ECONOMY IN THE LONG RUN

Lecture 2C A Classical Open Economy

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Supplement Readings

西方经济学

- (1) M14.1; S11.1, S11.2. ¹
- (2) 其他资料: [GM] THE OPEN ECONOMY



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

西方经济学

- (1) 掌握小国经济模型。
- (2) 掌握大国经济模型。
- (3) 领会马工程教材精神。



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Outline

1 Exchange Rates

- Exchange Rates
- The Law of One Price
- Purchasing Power Parity
- Net Exports
- Uncovered Interest Parity
- Assumptions

2 Case: Small Home vs. Large Foreign

3 Case: Large Home vs. Large Foreign

4 马工程教材疑难重点

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Exchange Rates

$e \downarrow$ $\epsilon \downarrow$ 代表本国货币升值, 对应可购买物变多

Definition 1

名义汇率 e

实际汇率 ϵ

The nominal exchange rate, denoted by e , is the price of the domestic currency in terms of foreign currency. The real exchange rate, denoted by ϵ , is the price of domestic goods in terms of goods of a foreign country. ϵ can be written as $\epsilon = e \times P/P^*$, where P denotes the price of domestic goods and P^* the price of goods of a foreign country.

马工程观点 (M, 2019, p.182)

名义汇率是两个国家（或地区）货币的相对价格，即一种货币能兑换另一种货币的数量，用 e 表示。名义汇率有两种不同的标价方法。一种是直接标价法，是用本国货币形式表示的国外货币的价格，即购买 1 单位或 100 单位的外币应该付出多少本国货币，故又称为应付标价法。这样，当本币升值时，购买单位外币所必须支付的本币就减少，即本币升值，汇率下降，也就是说，汇率升降与本国货币对外价值的高低成反比。另一种为间接标价法，即用国外货币表示本国货币的价格。它以购买一定单位（如 1 单位）的本国货币为标准来计算应收多少单位外币，故又称为应收标价法。由于这种标价法将本国货币设定为一定的数额，当本币币值上升时，单位本币所能兑换的外币就增加，也就是说，外币价值的高低和汇率的升降成反比。为了方便起见，本章采用间接标价法。

$$\epsilon = e \times \frac{P}{P^*}$$

国内商品价格
国外商品价格

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(第二版) 上册

《西方经济学》编写组



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The Law of One Price

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Proposition 2 (The Law of One Price)

If markets are perfectly competitive ^{完全竞争市场} and there are no frictions ^{无交易摩擦}, then identical goods sold in different locations must sell for the same price in terms of a common currency.



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Purchasing Power Parity

Proposition 3 (Purchasing Power of Parity [PPP])

*If markets are perfectly competitive and there are no frictions, then the real exchange rate ϵ is equal to 1.*¹ 绝对购买力平价

Proposition 3 implies $e = P^*/P$, which is called absolute PPP. In form of growth rate, $\frac{\dot{e}}{e} = \frac{\dot{P}^*}{P^*} - \frac{\dot{P}}{P}$, which is called relative PPP.

增长率形式表示

绝对 PPP: 实际汇率 $\epsilon = 1$. 货币购买力相同

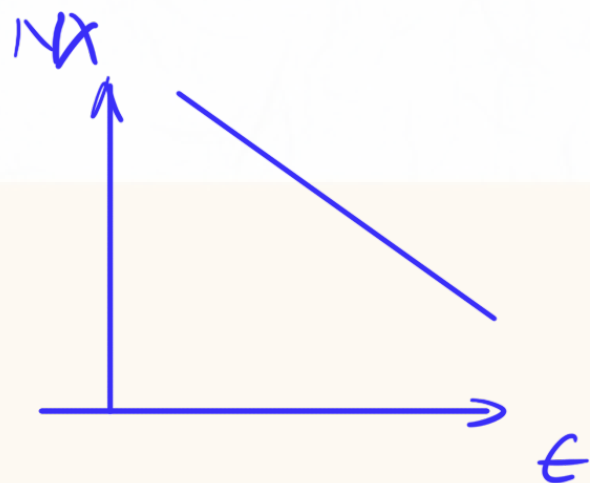
相对 PPP: 名义利率变动率 = 两国通货膨胀率之差

¹Parity: the quality or state of being equal or equivalent; equivalence of a commodity price expressed in one currency to its price expressed in another. —Merriam-Webster

Net Exports

If ϵ is high, foreign goods are relatively cheap and domestic goods are relatively expensive. Other things being equal, a higher real exchange rate implies that a country imports more and exports less. Therefore the net exports NX is decreasing in ϵ .

$$NX = NX(\epsilon), \quad NX'(\epsilon) < 0$$



Uncovered Interest Parity

UIP 无抛补利率平价

Proposition 4

完全流动

If capital mobility is perfect, then equilibrium in the foreign exchange market implies the uncovered interest parity:

无抛补利率平价

$$1 + \underset{\substack{\uparrow \\ \text{本国名义利率}}}{i_t} = (1 + \underset{\substack{\uparrow \\ \text{外国名义利率}}}{i_t^*}) \frac{e_t}{\mathbb{E}_t e_{t+1}}$$

or the uncovered real interest parity:

$$1 + \underset{\substack{\uparrow \\ \text{本国实际利率}}}{r_t} = (1 + \underset{\substack{\uparrow \\ \text{外国实际利率}}}{r_t^*}) \frac{\epsilon_t}{\mathbb{E}_t \epsilon_{t+1}}$$

→ 代入 $\epsilon_t = \frac{e_t P_t}{P_t^*}$ 即可

$$\frac{1 + i_t}{e_t} = \frac{1 + i_t^*}{e_{t+1}}$$



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Assumptions

小型开放经济 (忽略其对世界利率的影响)

Definition 5

大型开放经济 (计入其对世界利率的影响)

An economy is a small open economy if it is a small part of the world market and the net capital flow of the economy has only a negligible effect on the world interest rate. An economy is a large open economy if it is a large part of the world market and the net capital flow of the economy has a significant effect on the world interest rate.

Assumption 1 (Perfect Capital Mobility [PCM])

无成本进入国际金融市场

Residents of a country have full access to world financial markets.

Assumption 2 (Static Expectations of ϵ)

静态预期

$$\mathbb{E}_t \epsilon_{t+1} = \epsilon_t.$$

Assumption 3

PPP does not always hold.

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Case: Small Home vs. Large Foreign (Assumptions 1–3)

小型开放经济案例

Assumptions 1 and 2 imply $r = r^*$. For a small open economy, r^* is exogenous.

外定的. 不受本国影响

1. Market for Goods and Services

产品与服务市场

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(\overset{\text{不变}}{\downarrow} r^*) + \overset{\text{外生给定}}{\downarrow} \bar{G} + NX(\epsilon).$$

The only endogenous variable is $NX(\epsilon)$ which plays a role of equating the left-hand side and the right-hand side.

2. Market for Loanable Funds

可贷资金市场

$$\underset{\text{储蓄}}{\bar{Y} - C - G} = \bar{S} = I(r^*) + NX(\epsilon).$$

The difference between the supply and demand for loanable funds is called net foreign investment (NFI). 净外国投资 (对外投资)

3. Market for Foreign Exchange

外汇市场

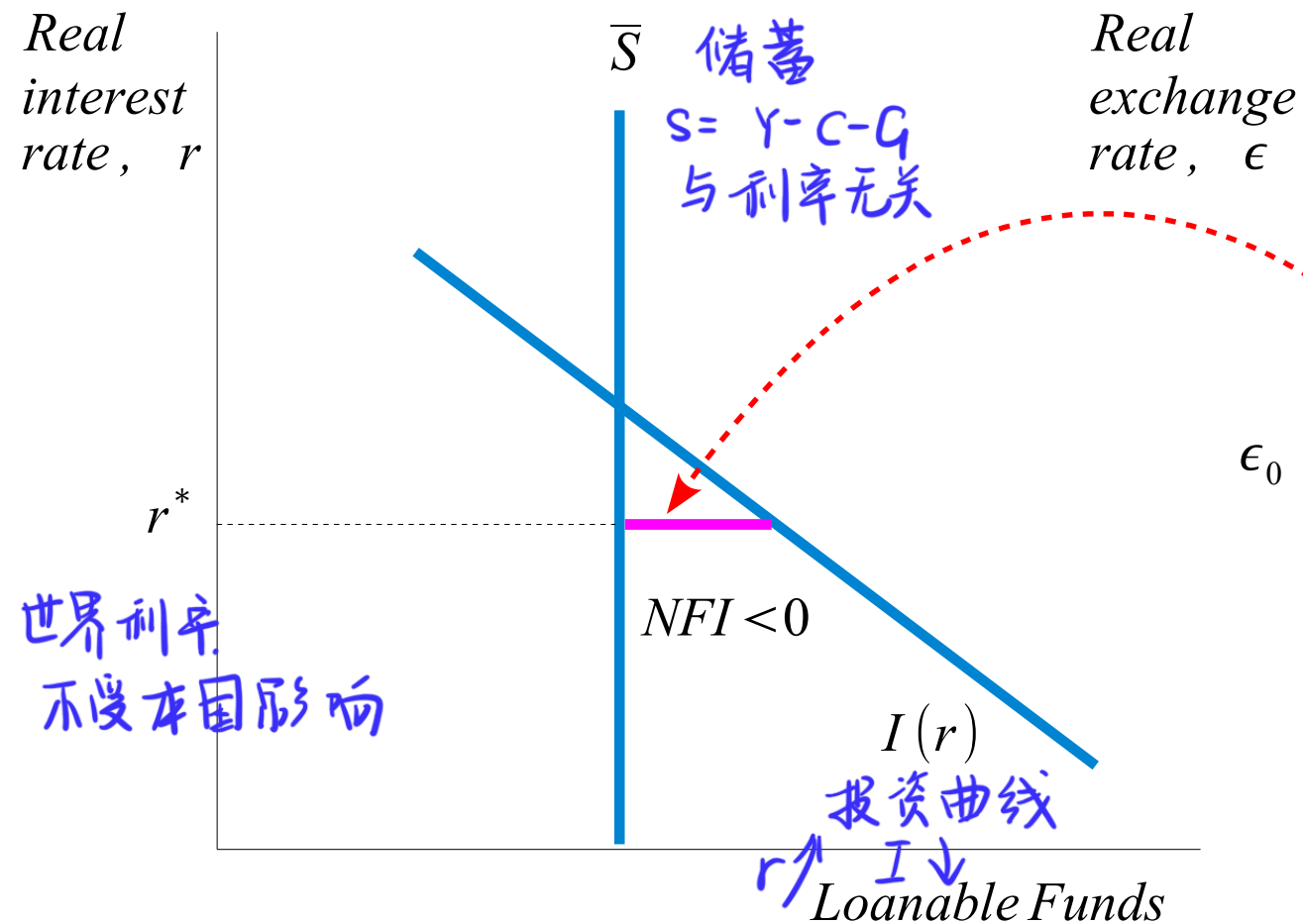
$$\bar{S} - I(r^*) = NX(\epsilon).$$

$\bar{S} - I(r^*)$ captures supply of domestic currency or demand for foreign currency; NX captures demand for domestic currency or supply of foreign currency.

The Complete Model for the Classical Small Open Economy with PCM and Floating ϵ

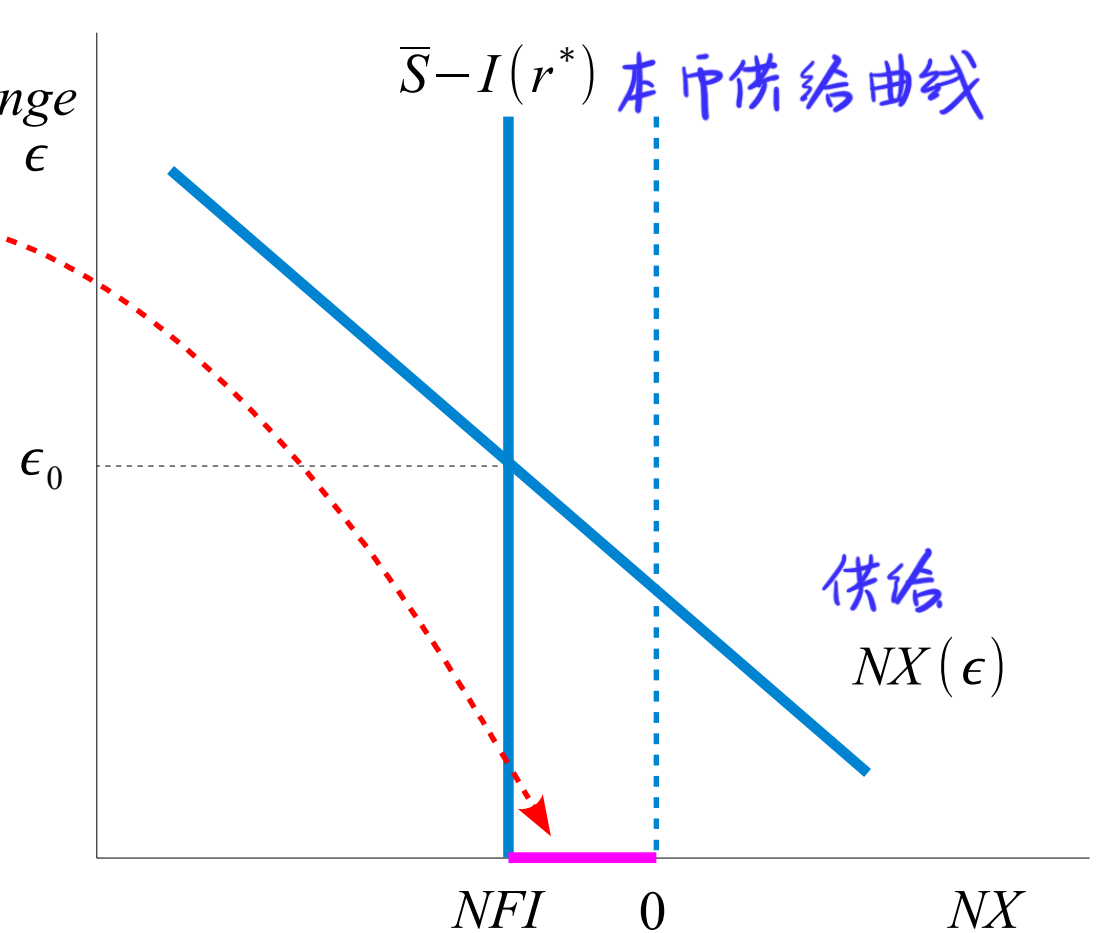
可贷资金市场

The Market for Loanable Funds



外汇市场

The Market for Foreign Exchange



可计算 $S-I$

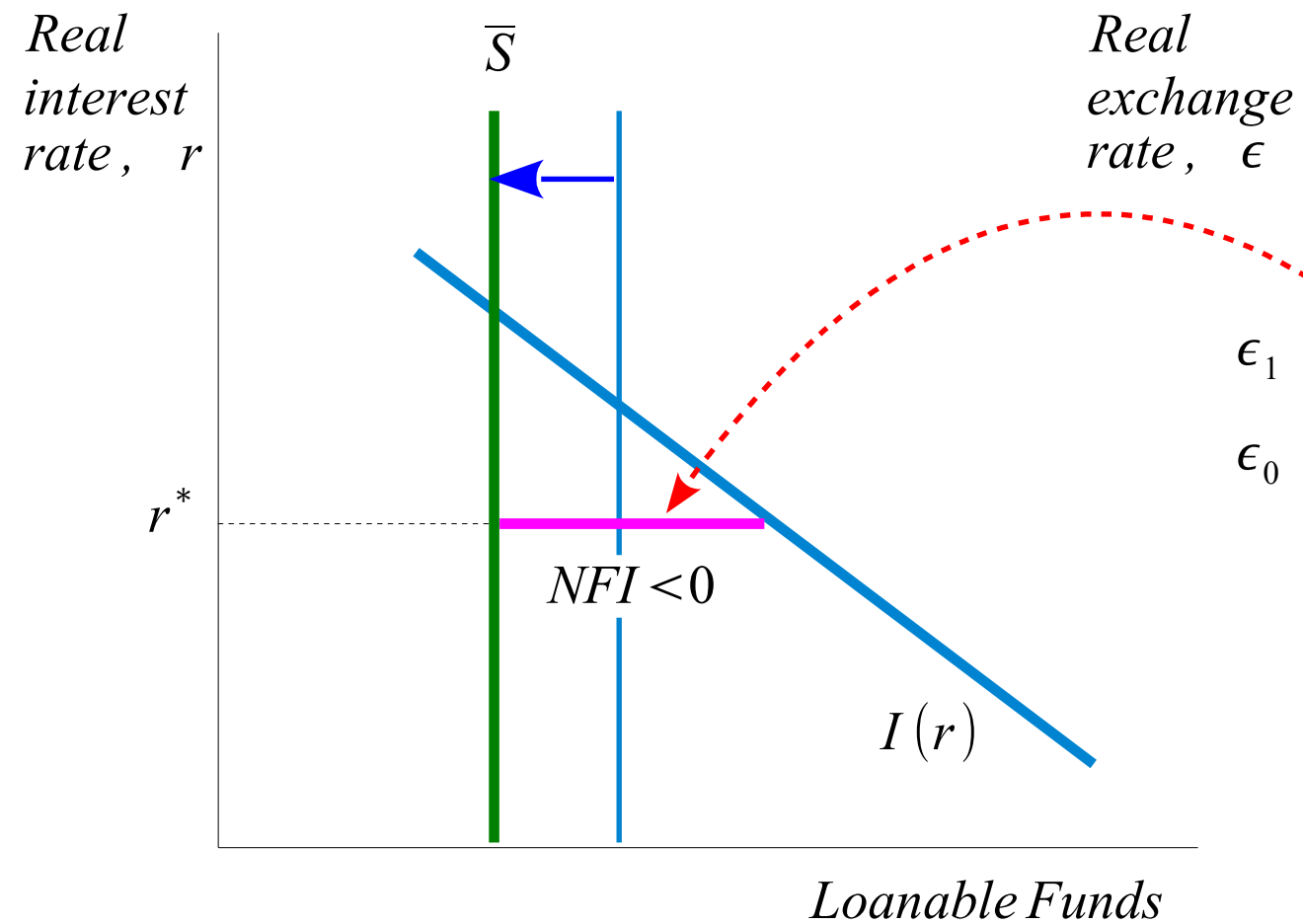
用 $S-I$ 计算 ϵ_0



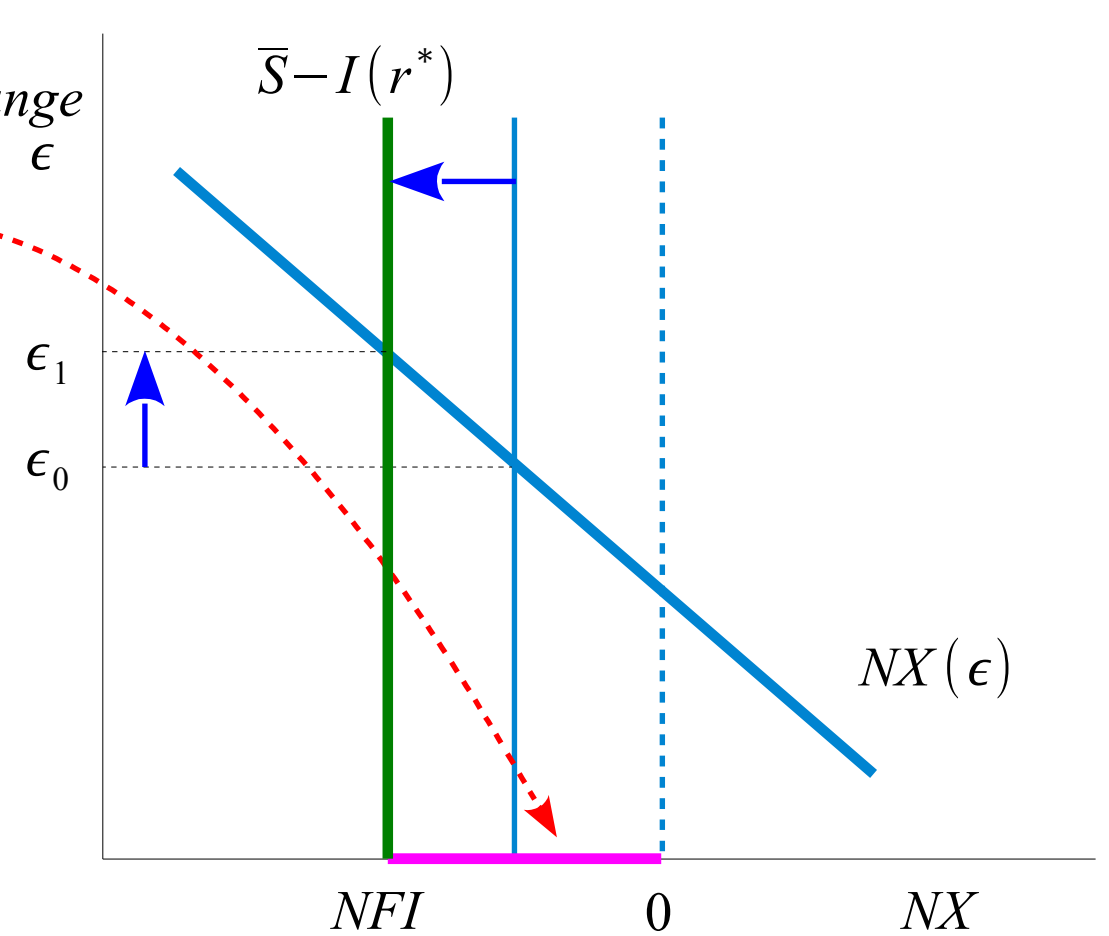
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A Reduction in Home's Saving

The Market for Loanable Funds



The Market for Foreign Exchange



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Case: Large Home vs. Large Foreign (Assumptions 1–3)

大型本国

大型外国

Large Home :

$$NX(\epsilon) = \left[\bar{Y} - \underbrace{C(\bar{Y} - \bar{T})}_{\text{国民储蓄}} - G \right] - I(r);$$

Large Foreign :

$$NX^*(1/\epsilon) = \left[\bar{Y}^* - C^*(\bar{Y}^* - \bar{T}^*) - G^* \right] - I^*(r^*);$$

Perfect Capital Mobility :

$$r = r^*;$$

The World is closed :

$$NX(\epsilon) + NX^*(1/\epsilon) \equiv 0.$$

It follows that

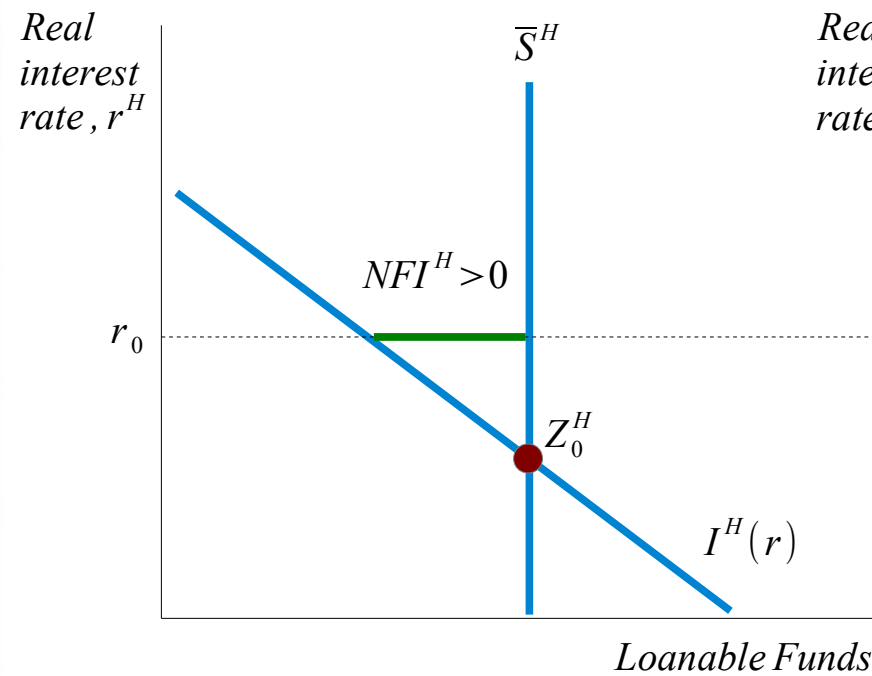
$$\overbrace{\bar{S} + \bar{S}^*}^{\text{世界储蓄}} = \underbrace{I(r) + I^*(r)}_{\text{世界投资}};$$

$$NX(\epsilon) = \bar{S} - I(r).$$

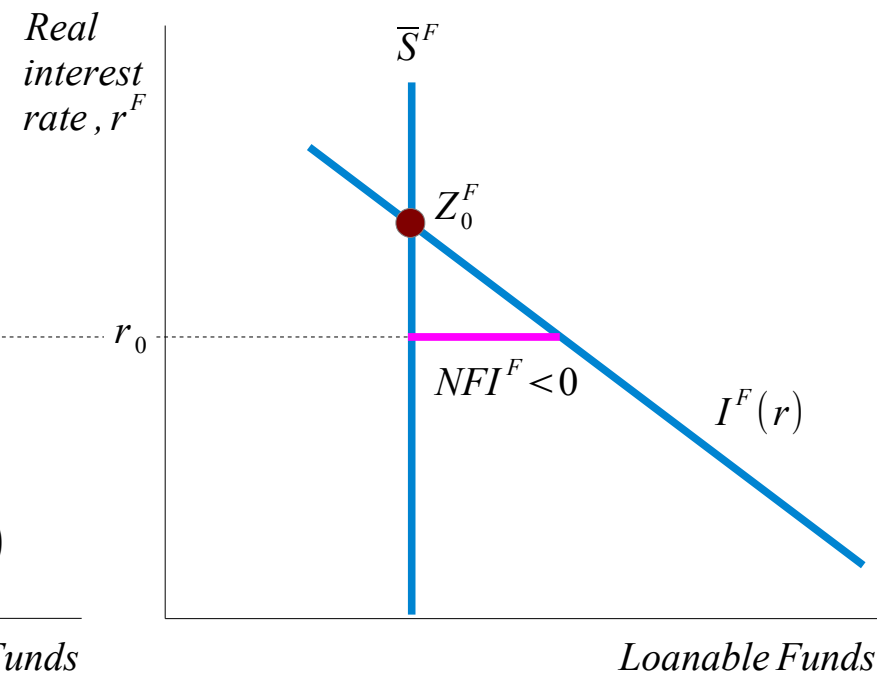


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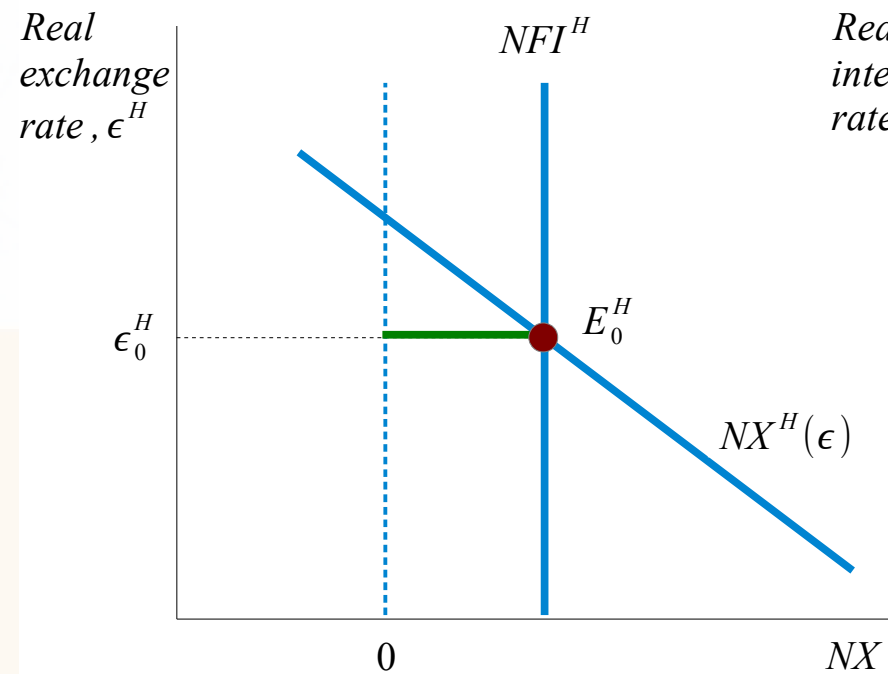
Case: Large Home vs. Large Foreign (Assumptions 1–3)



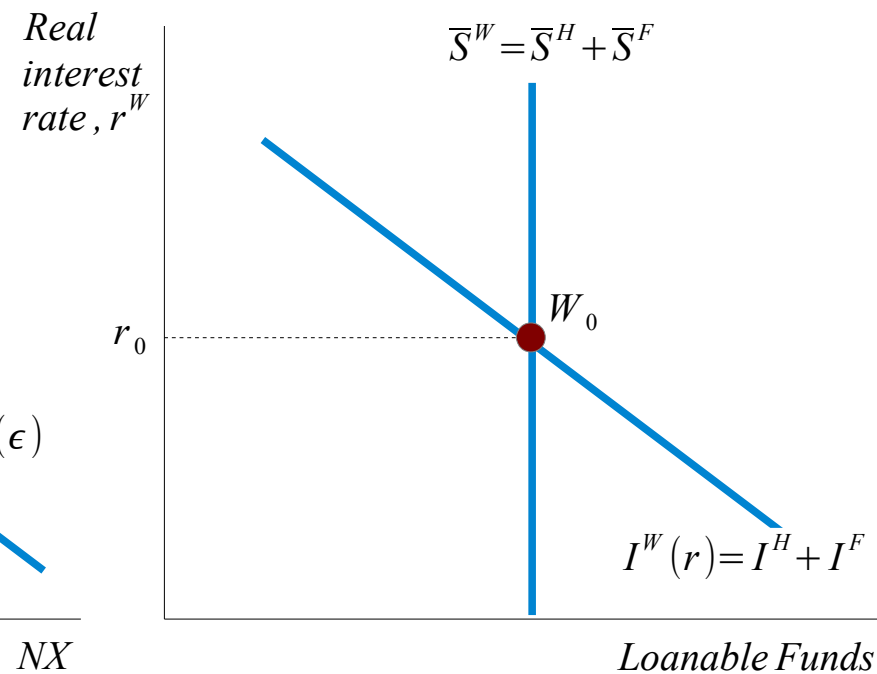
(a) Home Country



(b) Foreign Country

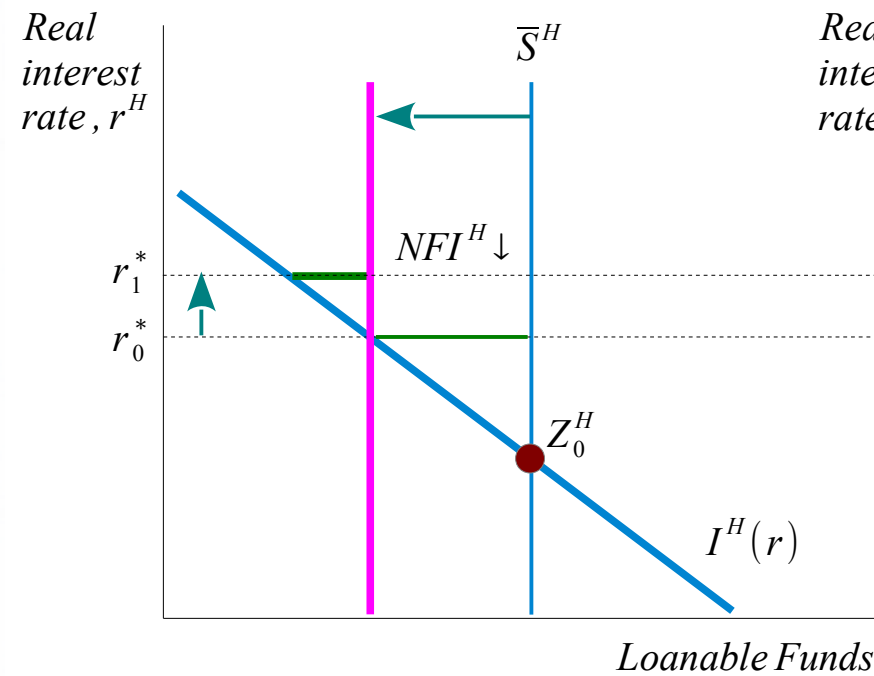


(c) Foreign Exchange Market

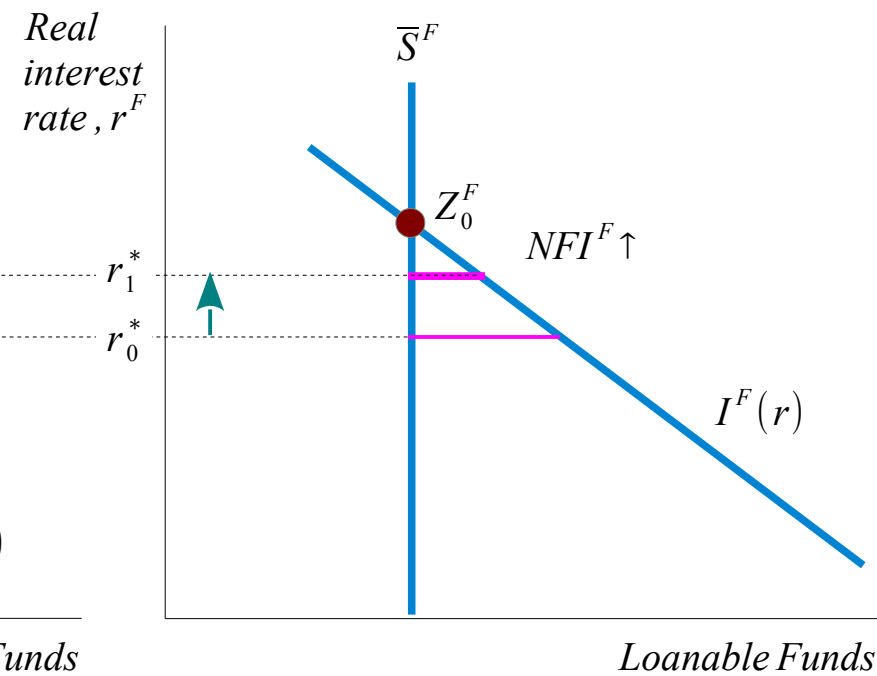


(d) The World

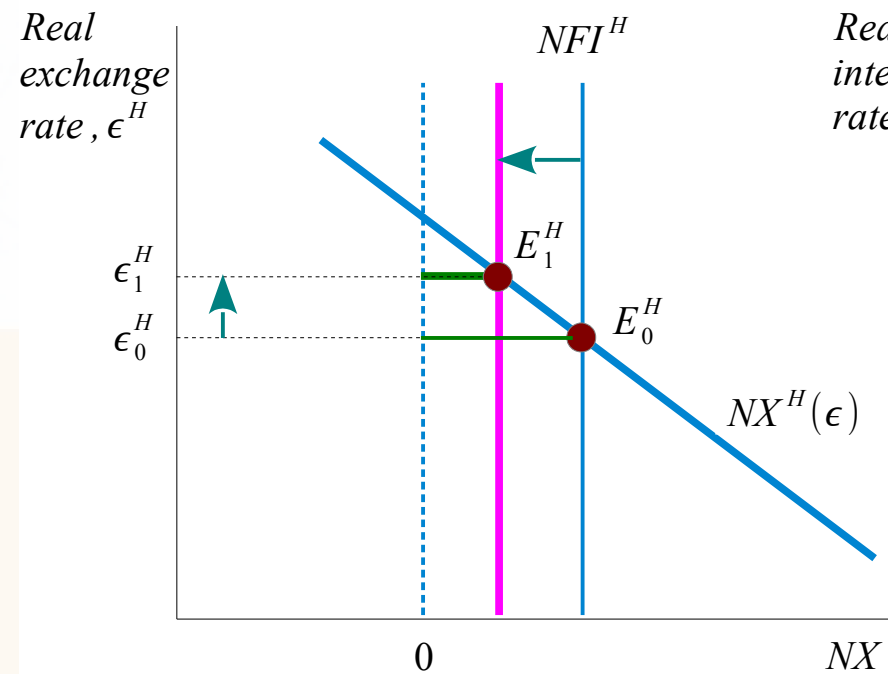
A Reduction in Home's Saving



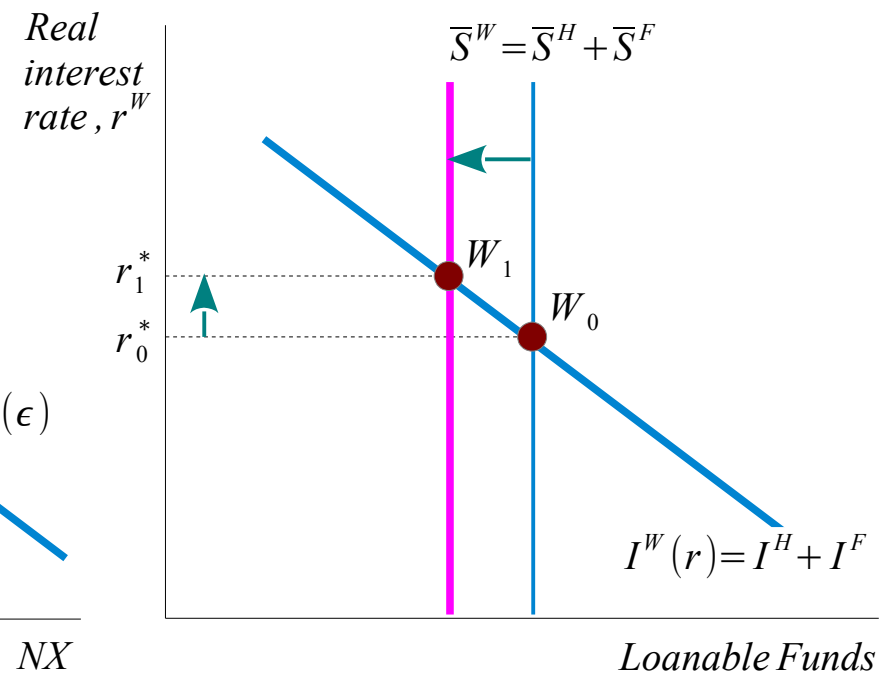
(a) Home Country



(b) Foreign Country



(c) Foreign Exchange Market



(d) The World

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西方经济学

- (1) 理解实际汇率的计算。
- (2) 掌握 The law of one price, PPP, uncovered interest parity.
- (3) 在 Small home-large foreign 模型中, $Y, C, I, S, NX, r, \epsilon$ 如何决定?
- (4) 在 Large home-large foreign 模型中, $Y, C, I, S, NX, r, \epsilon$ 如何决定?



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1 (E2, p.39)

根据马工程教材观点，为什么 1830 年后的经济学被马克思界定为“庸俗经济学”？

2 (E2, p.3)

根据马工程教材观点，应当如何评析西方经济学的科学因素和阶级属性？



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